

ATVB IN FOCUS:

Biological and Psycho-Socio-Environmental Determinants of Cardiovascular Disease

Series Editor: Daniel Sarpong

Financial Toxicity: Inequities and Impact on Cardiovascular Disease

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ABSTRACT: Healthcare costs are ever increasing, with impacts across all socioeconomic backgrounds. As a result, financial strain and financial toxicity are a consequences of rising healthcare costs and economic variations, with significant negative impacts on cardiovascular disease. This review explores financial toxicity and the interplay of net worth, financial mobility, and social determinants of health in cardiovascular disease while also addressing the role of advocacy and policy in public health. Given the significant multidimensional aspects of financial toxicity, understanding its impact on cardiovascular disease is crucial to improving outcomes on a population-wide scale.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: cardiovascular diseases ■ financial stress ■ policy ■ public health ■ socioeconomic factors

The cost of healthcare in the United States is growing significantly, with cardiovascular disease being an important contributor to that cost. In 2022, the country spent \$4.2 trillion on healthcare.¹ Cardiovascular disease is a leading cause of morbidity and mortality, with a high prevalence, considering that 1 in 3 adults received care for a cardiovascular risk factor or condition in 2020.¹ The decreasing rates of cardiovascular disease have plateaued since 2010 due to social determinants of health.² The cost of cardiovascular disease and stroke stretches beyond the direct healthcare costs, with indirect costs attributed to loss of productivity from morbidity and early mortality.¹ With accounting for inflation and population growth, healthcare expenses are expected to rise from \$393 billion in 2020 to \$1490 billion by 2050.² At the same time, productivity losses are projected to increase from \$234 billion to \$361 billion.² This results in a total economic burden of roughly \$1.851 trillion by 2050.¹ Similar trends of increasing productivity losses and catastrophic health expenditures have been noted worldwide, especially in low- and middle-income countries, where households are at a higher risk of financial strain due to cardiovascular care.³

Increased healthcare costs results in increased financial strain as a result of out-of-pocket expenses, including deductibles, copays, and coinsurance.⁴ This burden can contribute to harm for patients with decreased medication adherence for lifesaving therapeutics, especially in marginalized populations.⁵ Financial toxicity is the result of consequences from financial strain experienced by patients. Manifestations are often associated with deleterious impacts as a result of psychosocial distress, with negative impacts on a patient's quality of life and ability to receive care.⁶ Although the concept of financial toxicity initially emerged in the context of cancer care, it is now acknowledged as a significant concern in atherosclerotic cardiovascular disease. Patients often cope with lifelong medication, frequent hospital visits, and substantial cumulative expenses.⁴ In addition, financial mobility, which refers to the direction of change in a person's net worth and finances, also contributes to financial toxicity effects on cardiovascular disease. Given its importance, this

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Arterioscler Thromb Vasc Biol is available at www.ahajournals.org/journal/atvb

Nonstandard Abbreviations and Acronyms

CARDIA Coronary Artery Risk Development in Young Adults

review article aims to highlight the pathophysiology of financial toxicity as well as the relationship of income, net worth, and financial mobility on cardiovascular disease.

SOCIAL PATHOLOGY

There are several domains of financial toxicity that can be identified and used as targets for advocacy and policy change. These include both objective and subjective measures: objective measures relate to spending relative to affordability, whereas subjective measures pertain to patients' perceived difficulties with expenses.⁶ Subjective measures obtained by surveys were reported to be substantially greater than objective ones.

Beyond household spending measures, financial toxicity also leads to broader economic consequences, categorized as either direct medical costs (hospitalizations, medications, and procedures), direct nonmedical costs (transportation, informal caregiving, and home modifications), or indirect costs, such as productivity losses from illness and premature death. Neglecting these aspects underestimates the true burden of cardiovascular disease, as indirect costs can equal or exceed direct healthcare expenditures in chronic conditions. Recognizing these broader implications places financial toxicity not only as a challenge for patients but also as a societal economic burden, affecting cost-effectiveness analyses, budget impact assessments, and ultimately influencing policy decisions regarding the financing of cardiovascular care.

Although financial toxicity in atherosclerotic cardiovascular disease can affect all families, it is most prevalent in low-income families, uninsured patients, and nonelderly patients who are not covered by Medicare.⁶ In some cases, healthcare spending for low-income families may exceed 20% of their annual income⁶ while also making 5-fold lower than that of middle- and high-income families.⁶ Subjectively, more than half of low-income families reported challenges paying bills and had 2-fold higher odds of being unable to pay their bills. This aligns with the World Health Organization's framework for catastrophic health expenditure,⁷ which defines financial catastrophe as out-of-pocket spending exceeding 10% of household income or consumption, or 25% of nonfood expenditures.⁷ Such disproportionate spending not only exacerbates immediate financial hardship but also heightens the likelihood of long-term impoverishment and intergenerational impacts, thereby reinforcing the cycle of poverty and illness in affected households.

Uninsured patients are more likely to have worse outcomes and experience more financial toxicity. Even with the increased access to health insurance following the passage of the Affordable Care Act, 1 in 4 Americans

Highlights

- Financial toxicity remains an underrecognized but major determinant of cardiovascular health, with rising healthcare costs, out-of-pocket expenses, and economic instability contributing to worse cardiovascular outcomes through medication nonadherence, psychosocial distress, food insecurity, and reduced quality of life.
- Socioeconomic determinants involving structural inequalities and environmental resources matter beyond income alone. Insurance status, neighborhood disadvantages, food deserts, and unequal access to resources interact with financial toxicity to sustain cardiovascular disparities.
- Reducing cardiovascular inequities requires multidisciplinary and system-level solutions with public policy, community interventions, equitable healthcare financing, and social determinant screening.

still delay treatment due to cost despite having access to insurance.⁶ Insurance type also affects healthcare costs, as families with private insurance are more prone to catastrophic health expenditures, with 1 in 3 facing this issue, compared with 1 in 5 for those with public insurance.⁶ Private nonemployment-based insurance had 1.5-fold higher spending on premiums compared with employment-based and 10-fold higher than public.⁶ Nonelderly patients have higher rates of financial toxicity due to lack of Medicare coverage, more work responsibilities, and financial responsibilities for dependents.⁶ Younger adults (18–39 years old) have higher financial toxicity compared with middle-aged adults despite lower rates of atherosclerotic cardiovascular disease due to limited insurance and financial reserve.⁶

Financial toxicity has multiple downstream consequences that negatively affect patient outcomes, including psychosocial distress and indirect challenges related to healthcare costs, such as food insecurity. Psychosocial distress from healthcare-related finances can cause worsened perception of health, mental health disorders, and lower quality of life. Subjective financial toxicity is also an important contributor to medication noncompliance. Patients with difficulty paying bills also delay seeking medical care regardless of insurance status at a rate 2 to 3× higher than those who have no difficulty paying medical bills.⁶ Beyond health outcomes, these consequences incur indirect costs through productivity losses. Finally, food insecurity is a late downstream effect of financial toxicity that affects the entire family and is associated with a higher 10-year risk for cardiovascular disease (Figure).⁶

INCOME AND CARDIOVASCULAR DISEASES

Income refers to resources entering a household and has been examined for its effects on the socioeconomic aspects of health. A common consideration has been

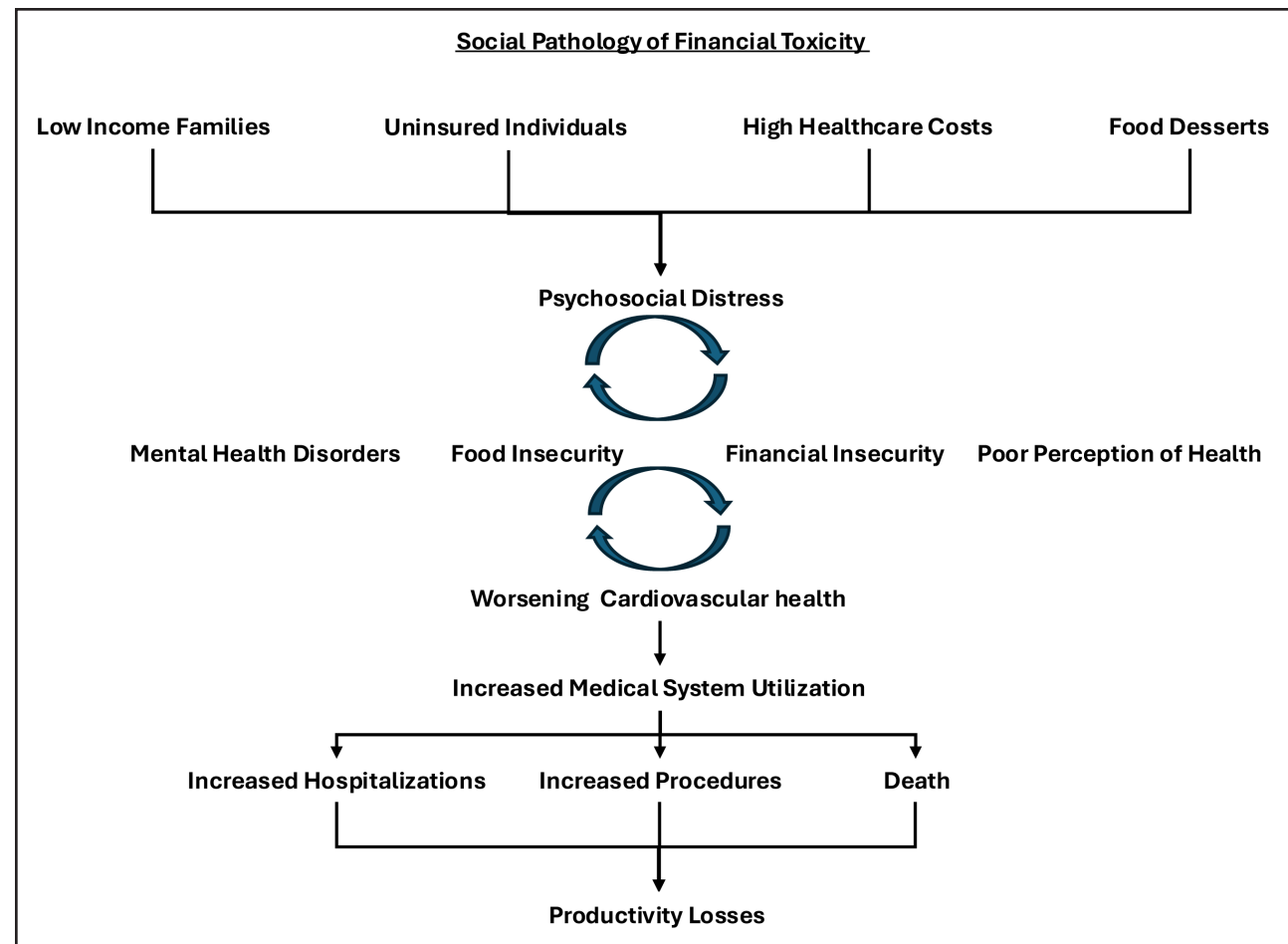


Figure. Financial toxicity as a mediator of cardiovascular health disparities.

Financial toxicity represents the adverse financial consequences of healthcare costs and socioeconomic inequities. Low income, lack of insurance, and high healthcare expenditures contribute to financial strain and reduced financial mobility. This social pathology promotes psychological stress, food insecurity, and barriers to preventive and therapeutic care, which may accelerate cardiovascular disease progression through increased risk factor burden, delayed treatment, higher healthcare utilization, and adverse clinical outcomes.

that lower socioeconomic status contributes in a deleterious manner toward health-related outcomes. Although Hajat et al⁸ provided a more complex analysis of income's impact, illustrating that while wealth is positively associated with smoking initiation, it is also inversely related to obesity and hypertension. Similar patterns were found in a Canadian survey involving over 27 000 individuals, where Lemstra et al⁹ showed that household income was strongly and independently linked to cardiovascular disease, with hypertension being the primary condition influencing this association. Notably, the connection between low income and cardiovascular disease risk remained significant even after accounting for traditional clinical risk factors, highlighting the independent impact of socioeconomic conditions on cardiovascular outcomes.

In addition, the association of race, income, and cardiovascular disease among minority populations has also been taken into consideration and evaluated. In Lopez-Cevallos et al,¹⁰ a study involving 4971 Hispanic and Latino individuals further addressed this question, emphasizing

that wealth was initially associated with hypercholesterolemia and hypertension. Furthermore, these associations disappeared after socioeconomic adjustment, leaving only subgroup differences in obesity, such as those between Puerto Ricans and Central Americans.

Another feature of income's effect on cardiovascular disease involves the concept of family income-to-poverty ratio.¹¹ Calculated by dividing an individual's or family's income by poverty guidelines specific to the survey year, it is utilized as a measure of socioeconomic status. Validated in its ability to assess income disparity, several studies examined poverty ratio with cardiovascular risk and noted an inverse relationship, with lower poverty ratio accounting for a relatively higher degree of cardiovascular risk factors. A retrospective study conducted by Minhas et al¹¹ involving 35 932 individuals from the National Health and Nutrition Examination Survey observed a relationship where low income and low poverty ratio were associated with a higher prevalence of coronary artery disease, congestive heart failure, and stroke when adjusted for cofounders.

Beyond epidemiological associations, income and wealth also interact with the concept of financial toxicity, with increasing recognition of their negative impacts on cardiovascular disease. Financial toxicity refers to the financial strain experienced by patients as they access healthcare and has been considered to have deleterious impairment on those afflicted. Initially described among populations with cancer because of expenses associated with chemotherapy and other cancer-related care. Valero-Elizondo et al¹² studied financial toxicity and its relationship with atherosclerotic cardiovascular disease. They noted a higher proportion of financial toxicity associated with atherosclerotic cardiovascular disease compared with patients with cancer, with even worse financial toxicity in patients with both atherosclerotic cardiovascular disease and cancer.

The link between income and cardiovascular disease goes beyond simple epidemiological associations. Lower income restricts patients' ability to afford medications, adhere to long-term therapies, and access preventive services, leading to more avoidable hospitalizations and increased downstream costs.¹³ At the societal level, income disparities result in unequal productivity losses, emphasizing the cycle of poverty and illness.¹⁴ Thus, incorporating income-related inequities into cost-effectiveness and budget impact models is essential for creating interventions and policies that address not just clinical outcomes but also financial protection and equity in cardiovascular care.

NET WORTH AND CARDIOVASCULAR DISEASE

Financial net worth is defined as the resultant value of financial assets after removal of liabilities and may reflect the overall value an individual may attain. Similarly to income and wealth, net worth has also been another measure that has been investigated for its impact on cardiovascular diseases and outcomes, as it may portend an individual's financial stability and economic reserve. Spikes et al¹⁵ investigated the effects of self-reported net worth and its impact on blood pressure among Black women and found that those with negative net worth had significantly elevated blood pressure measurements in comparison to those with positive net worth.

Beyond static measures, net worth is also dynamic and may experience changes which can portend an effect on cardiovascular health. Negative wealth shocks are defined as large sudden losses involving net worth, which are seen in the setting of economic recessions and depressions, and may also provide clinically relevant cardiovascular effects. Pool et al¹⁶ investigated the effects of wealth loss and determined that sudden loss of wealth was associated with a higher mortality hazard ratio across the socioeconomic spectrum.

In addition, net worth is also another means by which an individual's or household's ability to withstand health shocks. Individuals with low or negative net worth are at greater risk of facing catastrophic health expenditures, medication nonadherence, and loss of access to care during economic downturns. Conversely, positive net worth can provide a buffer against these financial pressures, lowering the likelihood of cost-related nonadherence and negative health outcomes. Incorporating net worth into cardiovascular economic models could enhance our understanding of health disparities and the long-term sustainability of interventions, particularly in economically volatile situations.

FINANCIAL MOBILITY AND CARDIOVASCULAR DISEASE

Financial mobility refers to changes in a person's net worth and finances, with movement either positive or negative in direction. The effects of negative financial mobility, often associated with negative wealth shocks have been noted. Elfassy et al¹⁷ via the CARDIA study (Coronary Artery Risk Development in Young Adults) demonstrated strong associations of income volatility and income drops with the incidence of cardiovascular disease. In addition, there was a 2-fold increase in the risk of cardiovascular disease and death in those who experienced high volatility or >2 income drops over 15 years.

On the contrary, the effects of positive financial mobility have been demonstrated with improvement in cardiovascular outcomes. Machado et al¹⁸ demonstrated that upward financial mobility was associated with a decrease in cardiovascular risk even when individuals began in the lowest wealth group. In addition, those who experienced upward wealth mobility into the highest wealth group also experienced similar cardiovascular risk as those who started and remained in the highest wealth group.

In consideration of race, White participants experienced similar negative impacts on cardiovascular health with downward wealth mobility. Meanwhile, among Black and Hispanic participants who experienced downward wealth mobility, the negative cardiovascular outcomes were less evidenced. This was considered in part due to collider bias and barriers to care among Black and Hispanic participants, limiting the diagnosis of cardiovascular diseases in these populations.¹⁸

When considering race, they interestingly found that primarily White participants had the previously noted impacts on cardiovascular effects with downward wealth mobility. This was in comparison to Black and Hispanic participants who also experienced downward wealth mobility, where cardiovascular effects were less evident, which was believed to be due to collider bias and experienced barriers to care and being less likely to be diagnosed with cardiovascular disease.

Overall, the direction of financial mobility, regardless of upward or downward directions, has been shown to have association with either increased or decreased cardiovascular disease risk. Financial mobility serves as an important indicator of a household's resilience to economic and health shocks. Downward mobility exacerbates cardiovascular risk and heightens vulnerability to catastrophic health expenditures, medication nonadherence, and productivity losses, thus increasing the long-term societal costs associated with cardiovascular disease.¹⁹ Oppositely, upward financial mobility enhances financial security, lowers cost-related barriers to adherence, and can mitigate the indirect costs of illness. By incorporating financial mobility into economic models and policy evaluations, we can better identify populations at the greatest risk of adverse health outcomes and economic hardship, facilitating targeted interventions aimed at reducing disparities.

DEVELOPING ADVOCACY, POLICY, AND PUBLIC HEALTH APPROACHES

Social disadvantage has been shown to provide significant challenges toward income, net worth, and financial mobility, while also contributing toward financial toxicity, culminating in significant effects on cardiovascular disease. The development of advocacy, policy, and other public health programs are alternatives which may provide means to limit or improve the effects of these factors on cardiovascular diseases.

This can be seen in the case of environmental and community aspects through neighborhood planning. The characteristics of a neighborhood involving the presence of sidewalks, recreational spaces, as well as public transportation have also been tied to cardiovascular disease. Several studies²⁰ have linked residence in disadvantaged areas with increased incidence of cardiovascular disease independent of income, education, and employment, whereas neighborhoods with favorable characteristics that promoted physical activity showed improved cardiovascular risk factors.

In addition, addressing factors such as food desert²¹ remains an important objective for public health initiatives. Defined as areas with low income and low access to healthy foods, these areas are often affected by factors such as racism. Residents within these food deserts often rely on unhealthier options involving processed foods often retailed from fast food restaurants, convenience stores, and gas stations. As a result, a higher prevalence of cardiovascular disease is often present. Initiatives targeting these disparate areas have involved subsidization of healthy foods through utilization of the Supplemental Nutrition Assistance Program. Although other approaches may be warranted, Kelli et al²² also showed that cardiovascular disease was driven largely by individual income rather than lack of healthy foods in these areas.

Measures emphasized toward directing resources to supporting legislation and modification in the environment and neighborhoods are means toward addressing social disadvantage. Expanding pedestrian infrastructure and public safety contributes toward safer and healthier community environments. In addition, utilizing local infrastructure via schools, hospitals, and government offices allows for effective platforms for engagement and advocacy. Organizations such as the United States Preventive Services Task Force, which advocates and recommends behavioral counseling interventions for promoting healthy diets and physical activity among adults with cardiovascular risk factors, are examples of such efforts.

CURRENT CHALLENGES

Despite improvement in the incidence and treatment of cardiovascular disease,²³ the prevalence and mortality associated with cardiovascular disease have been noted to plateau. A large contribution to this discrepancy at the population level is in part due to inadequate and unequal distribution of improvement across the population.^{24,25} When stratified among socioeconomic groups, there was improvement in socioeconomic factors, although a discrepancy remained in the prevalence of cardiovascular disease. A decrease in prevalence occurred in that of the higher socioeconomic group, while lower socioeconomic groups faced either a plateau or increasing prevalence of cardiovascular disease.²⁶ In addition, another factor extraneous to income, net worth, and financial mobility involves racism and structural racism. Represented as situations such as discriminatory housing, biased policies, as well as implicit and explicit biases. Both in part represent chronic discrimination and are significant barriers toward the inequities associated with financial acquisition and mobility.

An additional challenge²⁷ is the limited and uneven integration of health technology assessment and cost-effectiveness evidence into cardiovascular decision-making worldwide. Examples such as the United Kingdom's National Institute for Health and Care Excellence, Canada's Canadian Agency for Drugs and Technologies in Health, and Australia's Pharmaceutical Benefits Advisory Committee use structured health technology assessment processes to guide resource allocation, while many health systems still lack explicit frameworks for evaluating the value of cardiovascular interventions. Even in systems with universal coverage, cardiovascular disease care often entails high out-of-pocket costs, variable adoption of cost-effective generics and biosimilars, and dependence on fragmented real-world data. These gaps lead to inefficient spending, ongoing inequities in access, and challenges in scaling preventive strategies, issues prevalent in both high-income and middle-income countries. Strengthening health technology assessment capacity and aligning cardiovascular care with value-based frameworks are critical global priorities.

FUTURE DIRECTIONS

Despite the overall improvement in incidence at the population level, there remains a further need to evaluate the factors and dynamics associated with income inequality. The development and utilization of multidisciplinary measures, such as public policy aimed at individual, community, and population levels, in addition to the creation of social support mechanisms against inequalities involving equity of income and financial mobilization, is necessary. In addition, the dismantling of structural racism and facilitation of equitable community planning and development is key in promoting equity among disparities in socioeconomic classes. Furthermore, the creation of social support mechanisms should be considered in regards to the concept of negative wealth shocks, particularly during periods of economic recessions and depression. Future research^{28,29} should broaden the application of economic evaluation and modeling to capture the long-term health and financial impacts of cardiovascular disease. Although quality-adjusted life years remain the primary outcome measure, emerging frameworks like productivity-adjusted life years and distributional cost-effectiveness analysis offer tools to evaluate both the economic effects of morbidity and the equity implications of various interventions. Integrating these methods into health technology assessment and policy development can help ensure that investments in cardiovascular care not only enhance survival and quality of life but also mitigate productivity losses and financial strain. Lastly, the identification of individuals in need and areas for improvement remains an important challenge. A unique opportunity involves the usage of Z-codes. Z-codes are diagnosis codes belonging to the *International Statistical Classification of Diseases, Tenth Revision*, which capture data pertinent to social determinants of health.³⁰ Usage and documentation with such codes would allow for collection of pertinent data that characterize social needs and discrepancies experienced by patients, allowing for tailored interventions. In addition, Z-codes could assist by adding further levels of complexity and in turn affect medical billing opportunities. Together, these approaches highlight interventions that provide the greatest value for disadvantaged populations and individuals, aligning efficiency with equity in future cardiovascular policy.

CONCLUSIONS

Despite significant research and treatments directed toward cardiovascular disease, it remains a leading cause of morbidity and mortality with increasing prevalence and a plateau in decreasing incidence. Financial variables involving increased healthcare costs, income, net worth, and financial mobility contribute to this discrepancy and

to an increased prevalence of cardiovascular disease. These financial factors with implications on cardiovascular disease remain an important challenge that requires a multidisciplinary approach with usage of public policy to allow for improvement in inequities associated with these financial variables. Future strategies should guarantee that interventions are assessed not only for clinical effectiveness but also for economic sustainability, productivity impact, and equity among socioeconomic groups. Utilizing tools such as cost-effectiveness analysis, productivity-adjusted life years, and distributional frameworks in policy and practice can help align cardiovascular care with both efficiency and fairness.

ARTICLE INFORMATION

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Sources of Funding

None.

Disclosures

None.

ATVB in Focus: Biological and Psycho-Socio-Environmental Determinants of Cardiovascular Disease. Series Editor: Daniel Sarpong

Imaging Findings Associated With Socioenvironmental Exposures Inform Mechanisms of Cardiovascular Disease

10.1161/ATVB.AHA.125.322470

Financial Toxicity: Inequities and Impact on Cardiovascular Disease

10.1161/ATVB.AHA.125.324015

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